United States Patent	12419024
Kind Code	B2
Date of Patent	September 16, 2025
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High density static random-access memory

Abstract

A semiconductor memory cell comprising six vertical-transport field-effect transistors (VTFET) on a wafer. The six VTFET are in a first layer. The six VTFET are in a first row.

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Appl. No.: **18/053451**

Filed: **November 08, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20240155822 A1	May. 09, 2024

Publication Classification

Int. Cl.: **H01L27/11** (20060101); **H01L23/48** (20060101); **H10B10/00** (20230101)

U.S. Cl.:

CPC **H10B10/12** (20230201); **H01L23/481** (20130101);

Field of Classification Search

CPC: G11C (11/412); H10B (10/00); H10B (10/12); H01L (23/481)

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Primary Examiner: Swanson; Walter H

Background/Summary

BACKGROUND OF THE INVENTION

(1) The present invention relates generally to the field of semiconductor device manufacture and more particularly a high density static random-access memory using vertical-transport field-effect transistors (VTFET).

(2) Semiconductor devices are fabricated by sequentially depositing insulating (dielectric) layers, conductive layers, and semiconductor layers of material over a semiconductor substrate, and

patterning the various layers using lithography to form circuit components and elements thereon. Generally, these semiconductor devices include a plurality of circuits which form an integrated circuit (IC) fabricated on a semiconductor substrate.

(3) A static random-access memory (SRAM) may be formed as a semiconductor device. An SRAM is a random-access memory that uses latching circuitry (flip-flop) to store each bit. A traditional SRAM is formed from six transistors. Two groups of two of the six transistors form two cross-couple inverters that store the bit in the cell. The storage cell has two stable states which are used to denote “0” or VSS and “1” or VDD. The fifth and sixth transistors are access transistors for reading/writing operations. Alternative SRAM configurations are realized that may use 4, 8, or 10 transistors.

(4) Current SRAM circuitry can take up large amounts of space on an integrated circuit. There is a need for a high density SRAM cell that reduces the size requirements and therefore increase the density of SRAM circuitry on an integrated circuit. VTFET, due to the flow of current vertically, has a smaller width and height compared to traditional planar field-effect transistors.

SUMMARY

(5) In a first embodiment, a semiconductor memory cell comprising six vertical-transport field-effect transistors (VTFET) is on a wafer. In the first embodiment, the six VTFET are in a first layer. In the first embodiment, the six VTFET are in a first row. In the first embodiment, power for two of the VTFET of the six VTFET is connected to a power delivery network on a backside of the wafer. In the first embodiment, ground for two of the VTFET of the six VTFET is connected to a power delivery network on a backside of the wafer. In the first embodiment, a bitline and a bitline bar for two of the VTFET of the six VTFET is connected to a power delivery network on a backside of the wafer and wherein a wordline for two of the VTFET of the six VTFET is connected to a power delivery network on a frontside of the wafer. In the first embodiment, a bitline and a bitline bar for two of the VTFET of the six VTFET is connected to a power delivery network on a frontside of the wafer and wherein a wordline for two of the VTFET of the six VTFET is connected to a power delivery network on a backside of the wafer. In the first embodiment, a first metal wire electrically connects a source/drain region of a first VTFET, a source/drain region of a second VTFET, a source/drain region of a third VTFET, a gate region of a fourth VTFET, and a gate region of a fifth VTFET. In a first embodiment, a second metal wire electrically connects a gate region of the second VTFET, a gate region of the third VTFET, a source/drain region of the fourth VTFET, a source/drain region of the fifth VTFET, and a source/drain region of the sixth VTFET. In the first embodiment, the first metal wire is within the semiconductor memory cell. In the first embodiment, the second metal wire is within the semiconductor memory cell. In the first embodiment, a fin end and a gate end of one or more VTFET of the plurality of VTFET are aligned to an RX edge.

(6) In a second embodiment, a plurality of vertical-transport field-effect transistors (VTFET) on a wafer. In the second embodiment, each VTFET of the plurality of VTFET are in a first layer. In the second embodiment, six VTFET of the plurality of VTFET form the memory cell. In the second embodiment, at least one memory cell is in a row adjacent to at least one other memory cell. In the second embodiment, power for two of the VTFET of the six VTFET is connected to a power delivery network on a backside of the wafer. In the second embodiment, ground for two of the VTFET of the six VTFET is connected to a power delivery network on a backside of the wafer. In the second embodiment, a bitline and a bitline bar for two of the VTFET of the six VTFET is connected to a power delivery network on a backside of the wafer and wherein a wordline for two of the VTFET of the six VTFET is connected to a power delivery network on a frontside of the wafer. In the second embodiment, a bitline and a bitline bar for two of the VTFET of the six VTFET is connected to a power delivery network on a frontside of the wafer and wherein a wordline for two of the VTFET of the six VTFET is connected to a power delivery network on a backside of the wafer. In the second embodiment, a first metal wire electrically connects a source/drain region of a first VTFET, a source/drain region of a second VTFET, a source/drain

region of a third VTFET, a gate region of a fourth VTFET, and a gate region of a fifth VTFET. In a second embodiment, a second metal wire electrically connects a gate region of the second VTFET, a gate region of the third VTFET, a source/drain region of the fourth VTFET, a source/drain region of the fifth VTFET, and a source/drain region of the sixth VTFET. In the second embodiment, the first metal wire is within the semiconductor memory cell. In the second embodiment, the second metal wire is within the semiconductor memory cell.
(7) In a third embodiment, a semiconductor SRAM comprises six vertical-transport field-effect transistors (VTFET) on a wafer. In the third embodiment, the six VTFET are in a first layer. In the third embodiment, the six VTFET are in a first row.
(8) In the fourth embodiment, a semiconductor memory array comprises a plurality of vertical-transport field-effect transistors (VTFET) on a wafer. In the fourth embodiment, six VTFET of the plurality of VTFET are arranged a memory cell of one or more memory cells in a first layer on the wafer. In the fourth embodiment, each memory cell of the one or more memory cells are in arranged in a single row. In the fourth embodiment, each memory cell of the one or more memory cells share a first continuous bottom source/drain region for a first VTFET and a second VTFET in each memory cell. In the fourth embodiment, the first continuous bottom source/drain region is connected to a power delivery network on a backside of the wafer. In the fourth embodiment, each memory cell of the one or more memory cells share a second continuous bottom source/drain region for a third VTFET and a fourth VTFET in each memory cell. In the fourth embodiment, the second continuous bottom source/drain region is connected to a power delivery network on a backside of the wafer. In the fourth embodiment, each memory cell of the one or more memory cells share a third continuous bottom source/drain region for a fifth VTFET and a fourth continuous bottom source/drain region for a sixth VTFET in each memory cell. In the fourth embodiment, the third continuous bottom source/drain region is connected to a power delivery network on a backside of the wafer. In the fourth embodiment, the fourth continuous bottom source/drain region is connected to a power delivery network on a backside of the wafer
(9) In a fifth embodiment, a semiconductor memory array comprises one or more vertical transistors on a wafer. In the fifth embodiment, the one or more vertical transistors are arranged in one or more memory cells in a first layer. In the fifth embodiment, each memory cell of the one or more memory cells are in a single row.
(10) Embodiments of the present invention provide for high density cell layout of SRAM technologies utilizing VTFET. Embodiments of the present invention provide for an SRAM including six VTFET in a single row. Embodiments of the present invention provide for bitline connections on a first side of the SRAM and wordline connections on a second side of the SRAM. Embodiments of the present invention provide for a single line connecting pull down, pass gate, pull up, and cross couple. Embodiments of the present invention provide for fin end and gate end alignment. Embodiments of the present invention provide for any number of signals (e.g., clock, buss, I/O, power, ground, etc.) to be distributed or delivered to source/drain/gate regions of VTFET through a backside power delivery network.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The above and other aspects, features, and advantages of various embodiments of the present invention will be more apparent from the following description taken in conjunction with the accompanying drawings.
(2) FIG. **1** depicts a cross-section view of a VTFET semiconductor structure with a frontside contact for the top source/drain region, bottom source/drain region, and gate region, in accordance with a first embodiment of the present invention.

(3) FIG. **2** depicts a cross-section view of a VTFET semiconductor structure with a frontside contact for the top source/drain region and gate region and a backside contact for the bottom source/drain region, in accordance with a first embodiment of the present invention.
(4) FIG. **3** depicts a cross-section view of a VTFET semiconductor structure with a frontside contact for the top source/drain and a backside contact for the bottom source/drain region and gate region, in accordance with a first embodiment of the present invention.
(5) FIG. **4** depicts a circuit schematic of an SRAM in accordance with an embodiment of the present invention.
(6) FIG. **5** depicts a top view of two adjacent SRAM in accordance with an embodiment of the present invention.
(7) FIG. **6** depicts a top view of an SRAM cell in accordance with a first embodiment of the present invention.
(8) FIG. **7** depicts a cross-sectional view of section X of an SRAM cell in accordance with an embodiment of the present invention.
(9) FIG. **8** depicts a cross-sectional view of section Y of an SRAM cell in accordance with an embodiment of the present invention.

DETAILED DESCRIPTION

(10) Embodiments of the present invention recognize that vertical-transport field-effect transistors (VTFET) have vertical current flow. Embodiments of the present invention recognize that VTFET include a bottom source/drain region and a top source/drain region. Embodiments of the present invention recognize that the bottom source/drain region is closer to the backside of the VTFET (closer to the wafer) and that the top source/drain region is closer to the frontside of the VTFET (closer to the traditional interconnect wiring). Embodiments of the present invention recognize that an input will be to one source/drain region and the output will be to one source/drain region and therefore one of either the input or the output will be on the backside of the device and one of either the input or the output will be on the frontside of the device. Therefore, embodiments of the present invention recognize that VTFET transistors are suitable for use in SRAM technologies.
(11) Embodiments of the present invention provide for high density cell layout of SRAM technologies utilizing VTFET. Embodiments of the present invention provide for an SRAM including six VTFET in a single row. Embodiments of the present invention provide for bitline connections on a first side of the SRAM and wordline connections on a second side of the SRAM. Embodiments of the present invention provide for a single line connecting pull down, pass gate, pull up, and cross couple. Embodiments of the present invention provide for fin end and gate end alignment. Embodiments of the present invention provide for any number of signals (e.g., clock, buss, I/O, power, ground, etc.) to be distributed or delivered to source/drain/gate regions of VTFET through a backside power delivery network.
(12) Some embodiments will be described in more detail with reference to the accompanying drawings, in which the embodiments of the present disclosure have been illustrated. However, the present disclosure can be implemented in various manners, and thus should not be construed to be limited to the embodiments disclosed herein. Reference will now be made in detail to the embodiments of the present invention, examples of which are illustrated in the accompanying drawings, wherein like reference numerals refer to like elements throughout.
(13) The following presents a summary to provide a basic understanding of one or more embodiments of the disclosure. This summary is not intended to identify key or critical elements or delineate any scope of the particular embodiments or any scope of the claims. Its sole purpose is to present concepts in a simplified form as a prelude to the more detailed description that is presented later. It is to be understood that aspects of the present invention will be described in terms of a given illustrative architecture; however, other architectures, structures, substrate materials, process features and steps can be varied within the scope of aspects of the present invention.
(14) Detailed embodiments of the claimed structures and methods are disclosed herein. The method

steps described below do not form a complete process flow for manufacturing integrated circuits, such as semiconductor devices. The present embodiments can be practiced in conjunction with the integrated circuit fabrication techniques currently used in the art, for advanced semiconductor devices, and only so much of the commonly practiced process steps are included as are necessary for an understanding of the described embodiments. The figures represent cross-section portions of a portion of an advanced semiconductor device after fabrication and are not drawn to scale, but instead are drawn to illustrate the features of the described embodiments. Specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to variously employ the methods and structures of the present disclosure. In the description, details of well-known features and techniques may be omitted to avoid unnecessarily obscuring the presented embodiments.

(15) It will also be understood that when an element such as a layer, region or substrate is referred to as being "on" or "over" another element, it can be directly on the other element or intervening elements can also be present. In contrast, when an element is referred to as being "directly on" or "directly over" another element, there are no intervening elements present. It will also be understood that when an element is referred to as being "connected" or "coupled" to another element, it can be directly connected or coupled to the other element or intervening elements can be present. In contrast, when an element is referred to as being "directly connected" or "directly coupled" to another element, there are no intervening elements present.

(16) For purposes of the description hereinafter, the terms "upper," "lower," "right," "left," "vertical," "horizontal," "top", "bottom," and derivatives thereof shall relate to the disclosed structures and methods, as oriented in the drawing figures. The terms "overlying," "atop," "over," "on "positioned on," or "positioned atop" mean that a first element is present on a second element wherein intervening elements, such as an interface structure, may be present between the first element and the second element. The term "direct contact" means that a first element and a second element are connected without any intermediary conducting, insulating or semiconductor layers at the interface of the two elements.

(17) In the interest of not obscuring the presentation of the embodiments of the present invention, in the following detailed description, some of the processing steps, materials, or operations that are known in the art may have been combined for presentation and illustration purposes and in some instances may not have been described in detail. Additionally, for brevity and maintaining a focus on distinctive features of elements of the present invention, description of previously discussed materials, processes, and structures may not be repeated with regard to subsequent Figures. In other instances, some processing steps or operations that are known may not be described. It should be understood that the following description is rather focused on the distinctive features or elements of the various embodiments of the present invention.

(18) The present embodiments can include a design for an integrated circuit chip, which can be created in a graphical computer programming language and stored in a computer storage medium (such as a disk, tape, physical hard drive, or virtual hard drive such as in a storage access network). If the designer does not fabricate chips or the photolithographic masks used to fabricate chips, the designer can transmit the resulting design by physical means (e.g., by providing a copy of the storage medium storing the design) or electronically (e.g., through the Internet) to such entities, directly or indirectly. The stored design is then converted into the appropriate format (e.g., GDSII) for the fabrication of photolithographic masks, which typically include multiple copies of the chip design in question that are to be formed on a wafer. The photolithographic masks are utilized to define areas of the wafer (and/or the layers thereon) to be etched or otherwise processed.

(19) Methods as described herein can be used in the fabrication of integrated circuit chips. The resulting integrated circuit chips can be distributed by the fabricator in raw wafer form (that is, as a single wafer that has multiple unpackaged chips), as a bare die, or in a packaged form. In the latter case, the chip is mounted in a single chip package (such as a plastic carrier, with leads that are

affixed to a motherboard or other higher-level carrier) or in a multichip package (such as a ceramic carrier that has either or both surface interconnections or buried interconnections). In any case, the chip is then integrated with other chips, discrete circuit elements, and/or other signal processing devices as part of either (a) an intermediate product, such as a motherboard, or (b) an end product. The end product can be any product that includes integrated circuit chips, ranging from toys and other low-end applications to advanced computer products having a display, a keyboard or other input device, and a central processor.

(20) It should also be understood that material compounds will be described in terms of listed elements, e.g., SiGe. These compounds include different proportions of the elements within the compound, e.g., SiGe includes SixGe1-x where x is less than or equal to 1, etc. In addition, other elements can be included in the compound and still function in accordance with the present principles. The compounds with additional elements will be referred to herein as alloys.

(21) Reference in the specification to "one embodiment" or "an embodiment", as well as other variations thereof, means that a particular feature, structure, characteristic, and so forth described in connection with the embodiment is included in at least one embodiment. Thus, the appearances of the phrase "in one embodiment" or "in an embodiment", as well any other variations, appearing in various places throughout the specification are not necessarily all referring to the same embodiment.

(22) References in the specification to "one embodiment", "other embodiment", "another embodiment," "an embodiment," etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is understood that it is within the knowledge of one skilled in the art to affect such feature, structure or characteristic in connection with other embodiments whether or not explicitly described.

(23) It is to be appreciated that the use of any of the following "/", "and/or", and "at least one of", for example, in the cases of "A/B", "A and/or B" and "at least one of A and B", is intended to encompass the selection of the first listed option (A) only, or the selection of the second listed option (B) only, or the selection of both options (A and B). As a further example, in the cases of "A, B, and/or C" and "at least one of A, B, and C", such phrasing is intended to encompass the selection of the first listed option (A) only, or the selection of the second listed option (B) only, or the selection of the third listed option (C) only, or the selection of the first and the second listed options (A and B) only, or the selection of the first and third listed options (A and C) only, or the selection of the second and third listed options (B and C) only, or the selection of all three options (A and B and C). This can be extended, as readily apparent by one of ordinary skill in this and related arts, for as many items listed.

(24) The terminology used herein is for the purpose of describing particular embodiments only and is not tended to be limiting of example embodiments. As used herein, the singular forms "a," "an" and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms "comprises," "comprising," "includes" and/or "including," when used herein, specify the presence of stated features, integers, steps, operations, elements and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components and/or groups thereof.

(25) Spatially relative terms, such as "beneath," "below," "lower," "above," "upper," and the like, can be used herein for ease of description to describe one element's or feature's relationship to another element(s) or feature(s) as illustrated in the FIGS. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the FIGS. For example, if the device in the FIGS. is turned over, elements described as "below" or "beneath" other elements or features would then be oriented

“above” the other elements or features. Thus, the term “below” can encompass both an orientation of above and below. The device can be otherwise oriented (rotated 90 degrees or at other orientations and the spatially relative descriptors used herein can be interpreted accordingly. In addition, be understood that when a layer is referred to as being “between” two layers, it can be the only layer between the two layers, or one or more intervening layers cat also be present.
(26) It will be understood that, although the terms first, second, etc. can be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another element. Thus, a first element discussed below could be termed a second element without departing from the scope of the present concept.
(27) In general, the various processes used to form a semiconductor chip fall into four general categories, namely, film deposition, removal/etching, semiconductor doping, and patterning/lithography. Deposition is any process that grows, coats, or otherwise transfers a material onto the wafer. Available technologies include but are not limited to physical vapor deposition (“PVD”), chemical vapor deposition (“CVD”), electrochemical deposition (“ECD”), molecular beam epitaxy (“MBE”) and more recently, atomic layer deposition (“ALD”) among others. Another deposition technology is plasma enhanced chemical vapor deposition (“PECVD”), which is a process that uses the energy within the plasma to induce reactions at the wafer surface that would otherwise require higher temperatures associated with conventional CVD. Energetic ion bombardment during PECVD deposition can also improve the film's electrical and mechanical properties.
(28) Semiconductor lithography is the formation of three-dimensional relief images or patterns on the semiconductor substrate for subsequent transfer of the pattern to the substrate. In semiconductor lithography, the patterns are formed by a light sensitive polymer called a photoresist. The pattern created by lithography or photolithography typically are used to define or protect selected surfaces and portions of the semiconductor structure during subsequent etch processes.
(29) Removal is any process such as etching or chemical-mechanical planarization (“CMP”) that removes material from the wafer. Examples of etch processes include either wet (e.g., chemical) or dry etch processes. One example of a removal process or dry etch process is ion beam etching (“IBE”). In general, IBE (or milling) refers to a dry plasma etch method that utilizes a remote broad beam ion/plasma source to remove substrate material by physical inert gas and/or chemical reactive gas means. Like other dry plasma etch techniques, IBE has benefits such as etch rate, anisotropy, selectivity, uniformity, aspect ratio, and minimization of substrate damage. Another example of a dry etch process is reactive ion etching (“RIE”). In general, RIE uses chemically reactive plasma to remove material deposited on wafers. High-energy ions from the RIE plasma attack the wafer surface and react with the surface material(s) to remove the surface material(s).
(30) Deposition processes for the metal liners and sacrificial materials include, e.g., chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), or gas cluster ion beam (GCIB) deposition. CVD is a deposition process in which a deposited species is formed as a result of chemical reaction between gaseous reactants at greater than room temperature (e.g., from about 25° C. about 900° C.). The solid product of the reaction is deposited on the surface on which a film, coating, or layer of the solid product is to be formed. Variations of CVD processes include, but are not limited to, Atmospheric Pressure CVD (APCVD), Low Pressure CVD (LPCVD), Plasma Enhanced CVD (PECVD), and Metal-Organic CVD (MOCVD) and combinations thereof may also be employed. In alternative embodiments that use PVD, a sputtering apparatus may include direct-current diode systems, radio frequency sputtering, magnetron sputtering, or ionized metal plasma sputtering. In alternative embodiments that use ALD, chemical precursors react with the surface of a material one at a time to deposit a thin film on the surface. In alternative embodiments that use GCIB deposition, a high-pressure gas is allowed to expand in a vacuum, subsequently condensing into clusters. The clusters can be ionized and directed onto a surface, providing a highly anisotropic deposition.

(31) Vertical transport field-effect transistors (VTFETs) have become viable device options for scaling semiconductor devices (e.g., complementary metal oxide semiconductor (CMOS) devices) to 5 nanometer (nm) node and beyond. VTFET devices include one or more fin channels with source/drain regions at ends of the fin channels on the top and bottom sides of the fins. Current flows through the fin channels in a vertical direction (e.g., perpendicular to a substrate), for example, from a bottom source/drain region to a top source/drain region. Vertical transport architecture devices are designed to address the limitations of horizontal device architectures in terms of, for example, density, performance, power consumption, and integration by, for example, decoupling gate length from the contact gate pitch, providing a FiN-FET-equivalent density at a larger contacted poly pitch (CPP), and providing lower middle-of-line (MOL) resistance.
(32) In a first embodiment, FIG. **1** shows a VTFET **100** with contact **114**, contact **124**, and contact **134** connected directly to interconnect wiring and/or a power delivery network (not shown) on the frontside of VTFET **100**. In a second embodiment, FIG. **2** shows a VTFET **200** with contact **214** and contact **234** connected directly to interconnect wiring and/or a power delivery network (not shown) on the frontside of VTFET **200** and a contact **224** connected directly to interconnect wiring and/or a power delivery network (not shown) on the backside of VTFET **200**. In a third embodiment, FIG. **3** shows a VTFET **300** with contact **314** connected directly to interconnect wiring and/or a power delivery network (not shown) on the frontside of VTFET **300** and contact **324** and contact **334** connected directly to interconnect wiring and/or a power delivery network (not shown) on the backside of VTFET **300**.
(33) FIG. **1** is a cross-sectional view of a VTFET **100** formed on a bulk substrate **102**. The substrate **102** may be formed of any suitable semiconductor structure, including various silicon-containing materials including but not limited to silicon (Si), silicon germanium (SiGe), silicon germanium carbide (SiGeC), silicon carbide (SiC) and multi-layers thereof. Although silicon is the predominantly used semiconductor material in wafer fabrication, alternative semiconductor materials can be employed as additional layers, such as, but not limited to, germanium (Ge), gallium arsenide (GaAs), gallium nitride (GaN), SiGe, cadmium telluride (CdTe), zinc selenide (ZnSe), etc. In one illustrative embodiment, substrate **102** is silicon.
(34) VTFET **100** includes STI region **104** comprised of a dielectric material such as silicon oxide or silicon oxynitride, and is formed by methods known in the art. For example, in one illustrative embodiment, STI region **104** is a shallow trench isolation oxide layer.
(35) VTFET **100** includes top source/drain region **110** and bottom source/drain region **120** on either end of fin **130**. In an embodiment, top source/drain region **110** is formed between dielectric layer **170**. In an embodiment, bottom source/drain region **120** is formed in substrate **102** between shallow trench isolation region **104**. Top source/drain region **110** and bottom source/drain region **120** are formed by, for example, epitaxial growth processes. The epitaxially grown top source/drain region **110** and bottom source/drain region **120** can be in-situ doped, meaning dopants are incorporated into the epitaxy film during the epitaxy process. Other alternative doping techniques can be used, including but not limited to, for example, ion implantation, gas phase doping, plasma doping, plasma immersion ion implantation, cluster doping, infusion doping, liquid phase doping, solid phase doping, etc., and dopants may include, for example, an n-type dopant selected from a group of phosphorus (P), arsenic (As) and antimony (Sb), and a p-type dopant selected from a group of boron (B), gallium (Ga), indium (In), and thallium (Tl) at various concentrations. For example, in a non-limiting example, a dopant concentration range may be 1×10.sup.18/cm.sup.3 to 1×10.sup.21/cm.sup.3. According to an embodiment, the bottom source/drain region **120** can be boron doped SiGe for a p-type field-effect transistor (P-FET) or phosphorous doped silicon for an n-type field-effect transistor (N-FET). It is to be understood that the term “source/drain region” as used herein means that a given source/drain region can be either a source region or a drain region, depending on the application.
(36) Terms such as “epitaxial growth and/or deposition” and “epitaxially formed and/or grown”

refer to the growth of a semiconductor material on a deposition surface of a semiconductor material, in which the semiconductor material being grown has the same crystalline characteristics as the semiconductor material of the deposition surface. In an epitaxial deposition process, the chemical reactants provided by the source gases are controlled and the system parameters are set so that the depositing atoms arrive at the deposition surface of the semiconductor substrate with sufficient energy to move around on the surface and orient themselves to the crystal arrangement of the atoms of the deposition surface. Therefore, an epitaxial semiconductor material has the same crystalline characteristics as the deposition surface on which it is formed. For example, an epitaxial semiconductor material deposited on a 11001 crystal surface will take on a 11001 orientation. In some embodiments, epitaxial growth and/or deposition processes are selective to forming on a semiconductor surface, and do not deposit material on dielectric surfaces, such as silicon dioxide or silicon nitride surfaces.

(37) Examples of various epitaxial growth processes include, for example, rapid thermal chemical vapor deposition (RTCVD), low-energy plasma deposition (LEPD), ultra-high vacuum chemical vapor deposition (UHVCVD), atmospheric pressure chemical vapor deposition (APCVD) and molecular beam epitaxy (MBE). The temperature for an epitaxial deposition process can range from 500° C. to 900° C. Although higher temperature typically results in faster deposition, the faster deposition may result in crystal defects and film cracking.

(38) A number of different sources may be used for the epitaxial growth of the compressively strained layer. In some embodiments, a gas source for the deposition of epitaxial semiconductor material includes a silicon containing gas source, a germanium containing gas source, or a combination thereof. For example, an epitaxial silicon layer may be deposited from a silicon gas source including, but not necessarily limited to, silane, disilane, trisilane, tetrasilane, hexachlorodisilane, tetrachlorosilane, dichlorosilane, trichlorosilane, and combinations thereof. An epitaxial germanium layer can be deposited from a germanium gas source including, but not necessarily limited to, germane, digermane, halogermane, dichlorogermane, trichlorogermane, tetrachlorogermane and combinations thereof. While an epitaxial silicon germanium alloy layer can be formed utilizing a combination of such gas sources. Carrier gases like hydrogen, nitrogen, helium and argon can be used. After epi formation, drive-in anneals can be applied to move the dopants closer to the bottom of the fin channels.

(39) In an embodiment, as used herein, a “semiconductor fin” or fin **130** refers to a semiconductor material that includes a pair of vertical sidewalls that are parallel to each other. As used herein, a surface is “vertical” if there exists a vertical plane from which the surface does not deviate by more than three times the root mean square roughness of the surface. In an embodiment, each fin **130** has a height ranging from approximately 20 nm to approximately 200 nm, and a width ranging from approximately 5 nm to approximately 30 nm. Other heights and/or widths that are lesser than, or greater than, the ranges mentioned herein can also be used in the present application. Each fin **130** is spaced apart from its nearest neighboring fin **130** by a pitch ranging from approximately 20 nm to approximately 100 nm; the pitch is measured from one point, or reference surface, of one semiconductor fin to the exact same point, or reference surface, on a neighboring semiconductor fin. Also, the fins **130** are generally oriented parallel to each other. Although the present application describes and illustrates a single fin **108**, any number of fins with gate region surrounding them may be used and the fins may be any shape.

(40) The fin **130** may be formed of any suitable semiconductor structure, including various silicon-containing materials including but not limited to silicon (Si), silicon germanium (SiGe), silicon germanium carbide (SiGeC), silicon carbide (SiC) and multi-layers thereof. Although silicon is the predominantly used semiconductor material in wafer fabrication, alternative semiconductor materials can be employed as additional layers, such as, but not limited to, germanium (Ge), gallium arsenide (GaAs), gallium nitride (GaN), SiGe, cadmium telluride (CdTe), zinc selenide (ZnSe), etc. In one illustrative embodiment, fin **130** is silicon.

(41) In an embodiment, bottom spacer layer **140** is formed on the STI region **104** and bottom source/drain region **120**. In an embodiment, bottom spacer layer **140** is formed around fin **130**. Suitable material for bottom spacer layer **140** includes, for example, silicon boron nitride (SiBN), siliconborocarbonitride (SiBCN), silicon oxycarbonitride (SiOCN), SiN and SiO.sub.x. Bottom spacer layer **140** can be deposited using, for example, directional deposition techniques, such as a high-density plasma (HDP) deposition and gas cluster ion beam (GCIB) deposition. The directional deposition deposits the spacer material preferably on the exposed horizontal surfaces, but not on the lateral sidewalls. Alternatively, the bottom spacer layer **140** can be formed by overfilling the space with dielectric materials, followed by chemical mechanical planarization (CMP) and dielectric recess.
(42) In an embodiment, top spacer layer **160** is formed on the gate region between fin **130** and dielectric layer **170**. In an embodiment, top spacer layer **160** is formed around fin **130**. Suitable material for top spacer layer **160** includes, for example, silicon boron nitride (SiBN), siliconborocarbonitride (SiBCN), silicon oxycarbonitride (SiOCN), SiN and SiO.sub.x. Bottom spacer layer **140** can be deposited using, for example, directional deposition techniques, such as a high-density plasma (HDP) deposition and gas cluster ion beam (GCIB) deposition. The directional deposition deposits the spacer material preferably on the exposed horizontal surfaces, but not on the lateral sidewalls. Alternatively, the top spacer layer **160** can be formed by overfilling the space with dielectric materials, followed by chemical mechanical planarization (CMP) and dielectric recess.
(43) A gate region is formed on bottom spacer layer **140** and around fin **130**. In illustrative embodiments, gate region is deposited on bottom spacer layer **140** and around fin **130** employing, for example, ALD, CVD, RFCVD, plasma enhanced CVD (PECVD), physical vapor deposition (PVD), or molecular layer deposition (MLD). The gate region may include a gate dielectric layer **150** and a gate conductor layer **132**. The gate dielectric layer **150** may be formed of a high-k dielectric material. Examples of high-k materials include but are not limited to metal oxides such as HfO.sub.2, hafnium silicon oxide (Hf—Si—O), hafnium silicon oxynitride (HfSiON), lanthanum oxide (La.sub.2O.sub.3), lanthanum aluminum oxide (LaAlO.sub.3), zirconium oxide (ZrO.sub.2), zirconium silicon oxide, zirconium silicon oxynitride, tantalum oxide (Ta.sub.2O.sub.5), titanium oxide (TiO.sub.2), barium strontium titanium oxide, barium titanium oxide, strontium titanium oxide, yttrium oxide (Y.sub.2O.sub.3), aluminum oxide (Al.sub.2O.sub.3), lead scandium tantalum oxide, and lead zinc niobate. The high-k material may further include dopants such as lanthanum (La), aluminum (Al), and magnesium (Mg). The gate conductor layer **132** may include a metal gate or work function metal (WFM). The WFM for the gate conductor layer may be titanium (Ti), titanium nitride (TiN), tantalum (Ta), tantalum nitride (TaN), aluminum (Al), titanium aluminum (TiAl), titanium aluminum carbon (TiAlC), a combination of Ti and Al alloys, a stack which includes a barrier layer (e.g., of TiN, TaN, etc.) followed by one or more of the aforementioned WFM materials, etc. It should be appreciated that various other materials may be used for the gate conductor layer **132** as desired.
(44) In an embodiment, dielectric layer **170** may be composed of, for example, silicon oxide (SiO.sub.x), undoped silicate glass (USG), fluorosilicate glass (FSG), borophosphosilicate glass (BPSG), a spin-on low-κ dielectric layer, a chemical vapor deposition (CVD) low-κ dielectric layer or any combination thereof. As indicated above, the term "low-κ" as used herein refers to a material having a relative dielectric constant κ which is lower than that of silicon dioxide. In an embodiment, the dielectric layer **170** can be formed using a deposition technique including, for example, chemical vapor deposition (CVD), plasma enhanced chemical vapor deposition (PECVD), physical vapor deposition (PVD), atomic layer deposition (ALD), evaporation, spin-on coating, or sputtering.
(45) In an embodiment, top source/drain region **110**, bottom source drain region **120** and gate region are connected to interconnect wiring and/or a power delivery network (not shown) through

contact **114**, contact **124**, and contact **134**, respectively. In an embodiment, as shown in FIG. **1**, contact **114**, contact **124**, and contact **134** to are formed to directly connect to interconnect wiring and/or a power delivery network (not shown) on the frontside of the VTFET **100**. In an embodiment, contact **114**, contact **124**, and contact **134** may include any suitable conductive material, such as, for example, copper, aluminum, tungsten, cobalt, or alloys thereof. Examples of deposition techniques that can be used include, for example, chemical vapor deposition (CVD), plasma enhanced chemical vapor deposition (PECVD), or atomic layer deposition (ALD). In some cases, an electroplating technique can be used to form contact **114**, contact **124**, and contact **134**.
(46) In a second embodiment, FIG. **2** shows a VTFET **200** with contact **214** and contact **234** connected directly to interconnect wiring and/or a power delivery network (not shown) on the frontside of VTFET **200** and a contact **224** connected directly to interconnect wiring and/or a power delivery network (not shown) on the backside of VTFET **200**. In the second embodiment, VTFET **200** shares substantially similar features to the features described above in reference to VTFET **100**. For example, top source/drain region **210** is substantially similar to top source/drain region **110**. It should be noted, while a substrate, similar to substrate **102** shown in FIG. **1**, is not shown in FIG. **2**, it would be known to one skilled in the art that VTFET **200** would be formed on a substrate similar to substrate **102** shown in FIG. **1**.
(47) In the second embodiment, the orientation of contacts in VTFET **200** are the primary differences as compared to VTFET **100**. In the second embodiment, contact **214** and contact **234** are connected directly to interconnect wiring and/or a power delivery network (not shown) on the frontside of VTFET **200** and a contact **224** connected directly to interconnect wiring and/or a power delivery network (not shown) on the backside of VTFET **200**.
(48) In a third embodiment, FIG. **3** shows a VTFET **300** with contact **314** connected directly to interconnect wiring and/or a power delivery network (not shown) on the frontside of VTFET **300** and contact **324** and contact **334** connected directly to interconnect wiring and/or a power delivery network (not shown) on the backside of VTFET **300**. In the third embodiment, VTFET **300** shares substantially similar features to the features described above in reference to VTFET **100** and VTFET **200**. For example, top source/drain region **310** is substantially similar to top source/drain region **110** and top source/drain region **210**. It should be noted, while a substrate, similar to substrate **102** shown in FIG. **1**, is not shown in FIG. **3**, it would be known to one skilled in the art that VTFET **300** would be formed on a substrate similar to substrate **102** shown in FIG. **1**.
(49) In the third embodiment, the orientation of contacts in VTFET **300** are the are the primary differences as compared to VTFET **100** and VTFET **200**. In the third embodiment, contact **314** is connected directly to interconnect wiring and/or a power delivery network (not shown) on the frontside of VTFET **300** and contact **324** and contact **334** are connected directly to interconnect wiring and/or a power delivery network (not shown) on the backside of VTFET **300**.
(50) FIG. **4** depicts a circuit schematic of an SRAM **400** in accordance with an embodiment of the present invention. In an embodiment, SRAM **400** consists of six transistors, first transistor **410**, second transistor **412**, third transistor **420**, fourth transistor **422**, fifth transistor **430**, and sixth transistor **432**. As known in the art, this is also called a 6T SRAM cell. As described herein, each of the six transistors is a VTFET structure. However, embodiments of the present invention recognize that some or all of these six transistors maybe other types of transistors.
(51) In an embodiment, each bit in the SRAM **400** cell is stored on the third transistor **420**, fourth transistor **422**, fifth transistor **430**, and sixth transistor **432**. In an embodiment, the third transistor **420** and fourth transistor **422** form a first inverter. In an embodiment, the fifth transistor **430** and sixth transistor **432** form a second inverter. In an embodiment, the first inverter is cross-coupled with the second inverter. In other words, the output **440** of the first inverter is connected to the gate **446** of the second inverter and the output **442** of the second inverter is connected to the gate **444** of the second inverter. In an embodiment, the output **440** of the first inverter is also connected to a source/drain region of the first transistor **410**. In an embodiment, the output **442** of the second

inverter is connected to a source/drain region of the second transistor **412**. In an embodiment, a source/drain region of both the third transistor **420** and fifth transistor **430** is connected to a power or VDD supply **426** and **436**, respectively. In an embodiment, a source/drain region of both the fourth transistor **422** and sixth transistor **432** is connected a ground or GND supply **428** and **438**, respectively. In an embodiment, a gate region of the first transistor **410** and second transistor **412** is connected to a word line **450**. In an embodiment, word line **450** controls read and write operations in the SRAM **400** cell. In an embodiment, a source/drain region of the first transistor **410** is connected to bitline **460** and a source/drain region of the second transistor **412** is connected to bitline_bar **462**. In an embodiment, bitline **460** and bitline_bar **462** are used to transfer data for both read and write operations. It should be noted, an SRAM may only include a bitline **460** or bitline_bar **462** but generally both signals are provided to improve noise margins when reading and writing to the SRAM cell. As shown in FIG. **4**, the first transistor **410**, second transistor **412**, fourth transistor **422**, and sixth transistor **432** are “n-type” transistors and the third transistor **420**, and fifth transistor **430** are “p-type” but other configurations of doping may be realized.

(52) FIG. **5** depicts a top view of two adjacent SRAM **500** in accordance with an embodiment of the present invention. As shown in FIG. **5**, a first SRAM **502** cell is adjacent to a second SRAM **504** cell. For simplicity, only two cells are shown. However, one skilled in the art would recognize that any number of SRAM cell may be found adjacent to each other. It should be noted, the internal transistors for each SRAM cell are not shown in FIG. **5**, but as discussed above and below, each SRAM cell includes multiple transistors. In an embodiment, the first SRAM **502** cell is connected to a wordline (not shown) by wordline contacts **530**. In an embodiment, the second SRAM **504** cell is connected to a wordline (not shown) by wordline contacts **532**. In an embodiment, SRAM **500** includes bitline **510**, bitline_bar **512**, ground **520**, **522** or GND, and power **525** or VDD. In an embodiment, bitline **510**, bitline_bar **512**, ground **520**, ground **522**, and power **525** are found on the backside of the SRAM cell(s). In other words, the SRAM cell is on top of a wafer (not shown) and bitline **510**, bitline_bar **512**, ground **520**, ground **522**, and power **525** are between the SRAM cell(s) and the wafer. In an embodiment, wordline (not shown) is on the frontside of the SRAM cell(s) or on top of the SRAM cell(s). It should be noted, and discussed below, wordline and bitline/bitline_bar may be swapped in alternative configurations. It should be noted, and discussed below, VDD and GND may be swapped in alternative configurations.

(53) FIG. **6** depicts a top view of an SRAM **600** cell in accordance with a first embodiment of the present invention. In an embodiment, SRAM **600** cell includes a first transistor **630**, a second transistor **640**, a third transistor **650**, a fourth transistor **670**, a fifth transistor **660** and a sixth transistor **680**. It should be noted, first transistor **630** is similar to first transistor **410**, second transistor **640** is similar to second transistor **412**, third transistor **650** is similar to third transistor **420**, fourth transistor **670** is similar to fourth transistor **422**, fifth transistor **660** is similar to fifth transistor **430** and sixth transistor **680** is similar to sixth transistor **432**, as described in reference to function in FIG. **4**.

(54) As shown in FIG. **6**, SRAM **600** cell includes a backside power delivery network which includes power (i.e., VDD) **625**, ground (i.e., GND) **620**, **622**, bitline **610** and bitline_bar **612**. As discussed above, the backside power delivery network is found on the backside, or between the device layer (i.e., transistors) and the wafer the device layer is formed on. As shown in FIG. **6**, SRAM **600** cell includes a wordline **628**, **629** on either end of the SRAM **600** cell and connected to gate regions **632**, **642**, of the first transistor **630** and second transistor **640**, respectively. In an embodiment, the wordline **628**, **629** are formed on the frontside, or on top of the device layer and on the opposite side the wafer the device layer is formed on. In an embodiment, the wordlines **628**, **629** is within the traditional back end of the line interconnect wiring. As shown in FIG. **6**, wordlines **628**, **629** are on a first metal layer directly above the transistors of the SRAM **600** cell. In an alternative embodiment, wordline **628**, **629** are on any metal layer above the transistors of the SRAM **600** cell as long as the contact from the wordline **628**, **629** extends to the gate regions **632**,

642, of the first transistor **630** and second transistor **640**, respectively

(55) In an alternative embodiment, as noted above, power **625** may actually provide ground and ground **620**, **622** may provide power. In other words, the power and ground are swapped or flipped. In this alternative embodiment, one skilled in the art would recognize that the layout of a third transistor **650**, a fourth transistor **670**, a fifth transistor **660** and a sixth transistor **680** may be different. In an alternative embodiment, as noted above, wordline **628**, **629** may provide bitline **610** and bitline_bar **612** functionality and bitline **610**, bitline_bar **612** may provide wordline functionality. In other words, in this alternative embodiment, the wiring of the wordline is swapped or flipped with the wiring of bitline and bitline_bar. In this alternative embodiment, one skilled in the art would recognize that the layout of a first transistor **630**, a second transistor **640**, a third transistor **650**, a fourth transistor **670**, a fifth transistor **660** and a sixth transistor **680** may be different.

(56) In an embodiment, as shown in FIG. **6**, the first transistor **630** includes a bottom source/drain region (not shown), a fin **634**, a gate region **632**, and a top source/drain region **636**. In an embodiment, the bottom source/drain region (not shown) is connected to bitline **610** by a backside contact (not shown). In an embodiment, the gate region **632** is connected to wordline **628** through a frontside contact (not shown). In an embodiment, the top source/drain region **636** is connected to cross-couple **690** by a frontside contact **638**. In an embodiment, the bottom source/drain region (not shown) may be for the first transistor **630**. In an alternative embodiment, the bottom source/drain region (not shown) may be a shared bottom source/drain region that is also a shared bottom source/drain region of any number of first transistor(s) in an adjacent SRAM cell(s) (not shown). In an embodiment, as shown in FIG. **6**, the end of fin **634** and the end of the gate region **632** both extend and align on the RX edge of the first transistor **630** closest to the third transistor **650**. In an alternative embodiment, the end of fin **634** and the end of gate region **632** may not align with each other and/or may also not align on the RX edge of the first transistor **630**. For example, the edge of fin **634** and gate region **632** may align on a left edge of the first transistor **630** farthest from the third transistor **650**. In an embodiment, fin **634** has a first length, which is in the same direction as the longer length of SRAM **600** cell, and a second length which is shorter than the first length.

(57) In an embodiment, as shown in FIG. **6**, the second transistor **640** includes a bottom source/drain region (not shown), a fin **644**, a gate region **642**, and a top source/drain region **646**. In an embodiment, the bottom source/drain region (not shown) is connected to bitline_bar **612** by a backside contact (not shown). In an embodiment, the gate region **642** is connected to wordline **629** through a frontside contact (not shown). In an embodiment, the top source/drain region **646** is connected to cross-couple **694** by a frontside contact **648**. In an embodiment, the bottom source/drain region (not shown) may be for the first transistor **640**. In an alternative embodiment, the bottom source/drain region (not shown) may be a shared bottom source/drain region that is also a shared bottom source/drain region of any number of second transistor(s) in an adjacent SRAM cell(s) (not shown). In an embodiment, as shown in FIG. **6**, the end of fin **644** and the end of the gate region **642** both extend and align on the RX edge of the second transistor **640** closest to the fifth transistor **660**. In an alternative embodiment, the end of fin **644** and the end of gate region **642** may not align with each other and/or may also not align on the RX edge of the second transistor **640**. For example, the edge of fin **644** and gate region **642** may align on a right edge of the second transistor **640** farthest from the fifth transistor **660**. In an embodiment, fin **644** has a first length, which is in the same direction as the longer length of SRAM **600** cell, and a second length which is shorter than the first length.

(58) In an embodiment, as shown in FIG. **6**, the third transistor **650** includes a bottom source/drain region (not shown), a fin **654**, a shared gate region **652**/**672**, and a top source/drain region **656**. In an embodiment, the bottom source/drain region (not shown) is connected to ground **620** by a backside contact (not shown). In an embodiment, the shared gate region **652**/**672** is connected to cross-couple **694** by frontside contact **696** and the fourth transistor **670**, discussed below. In an

embodiment, shared gate region **652**/**672** is the gate region for both the third transistor **650** and fourth transistor **670**. In an embodiment, the top source/drain region **656** is connected to cross-couple **690** by a frontside contact **658**. In an embodiment, the bottom source/drain region (not shown) may be for the third transistor **650**. In an alternative embodiment, the bottom source/drain region (not shown) may be a shared bottom source/drain region that is also a shared bottom source/drain region of any number of third transistor(s) in an adjacent SRAM cell(s) (not shown). In an embodiment, as shown in FIG. **6**, the end of fin **654** and the end of the gate region **652**/**672** both extend and align on the RX edge of the third transistor **650** closest to the first transistor **630**. In an alternative embodiment, the end of fin **654** and the end of gate region **652**/**672** may not align with each other and/or may also not align on the RX edge of the third transistor **650**. For example, the edge of fin **654** may align on a right edge of the second transistor **640** farthest from the first transistor **630**. In an embodiment, fin **645** has a first length, which is in the same direction as the longer length of SRAM **600** cell, and a second length which is shorter than the first length.

(59) In an embodiment, as shown in FIG. **6**, the fifth transistor **660** includes a bottom source/drain region (not shown), a fin **664**, a shared gate region **662**/**682**, and a top source/drain region **666**. In an embodiment, the bottom source/drain region (not shown) is connected to ground **622** by a backside contact (not shown). In an embodiment, the shared gate region **662**/**682** is connected to cross-couple **690** by frontside contact **692** and the sixth transistor **680**, discussed below. In an embodiment, shared gate region **652**/**672** is the gate region for both the third transistor **650** and fourth transistor **670**. In an embodiment, the top source/drain region **666** is connected to cross-couple **694** by a frontside contact **668**. In an embodiment, the bottom source/drain region (not shown) may be for the fifth transistor **660**. In an alternative embodiment, the bottom source/drain region (not shown) may be a shared bottom source/drain region that is also a shared bottom source/drain region of any number of fifth transistor(s) in an adjacent SRAM cell(s) (not shown). In an embodiment, as shown in FIG. **6**, the end of fin **664** and the end of the gate region **662**/**682** both extend and align on the RX edge of the fifth transistor **660** closest to the second transistor **640**. In an alternative embodiment, the end of fin **664** and the end of gate region **662**/**682** may not align with each other and/or may also not align on the RX edge of the fifth transistor **660**. For example, the edge of fin **664** may align on a left edge of the fifth transistor **660** farthest from the second transistor **640**. In an embodiment, fin **664** has a first length, which is in the same direction as the longer length of SRAM **600** cell, and a second length which is shorter than the first length.

(60) In an embodiment, as shown in FIG. **6**, the fourth transistor **670** includes a bottom source/drain region (not shown), a fin **674**, a shared gate region **652**/**672**, and a top source/drain region **676**. In an embodiment, the bottom source/drain region (not shown) is connected to power **625** by a backside contact (not shown). In an embodiment, the shared gate region **652**/**672** is connected to cross-couple **694** by frontside contact **696** and the third transistor **650**, discussed above. In an embodiment, shared gate region **652**/**672** is the gate region for both the third transistor **650** and fourth transistor **670**, discussed above. In an embodiment, the top source/drain region **676** is connected to cross-couple **690** by a frontside contact **678**. In an embodiment, the bottom source/drain region (not shown) may be for the fourth transistor **670**. In an alternative embodiment, the bottom source/drain region (not shown) may be a shared bottom source/drain region that is also a shared bottom source/drain region of any number of fourth transistor(s) in an adjacent SRAM cell(s) (not shown). In an embodiment, as shown in FIG. **6**, the end of fin **674** and the end of the gate region **652**/**672** both extend and align on the RX edge of the fourth transistor **670** closest to the sixth transistor **680**. In an alternative embodiment, the end of fin **674** and the end of gate region **652**/**672** may not align with each other and/or may also not align on the RX edge of the fourth transistor **670**. For example, the edge of fin **674** and gate region **652**/**672** may align on a left edge of the fourth transistor **670** closest to the third transistor **650**. In an embodiment, fin **674** has a first length, which is in the same direction as the longer length of SRAM **600** cell, and a second length which is shorter than the first length.

(61) In an embodiment, as shown in FIG. **6**, the sixth transistor **680** includes a bottom source/drain region (not shown), a fin **684**, a shared gate region **662**/**682**, and a top source/drain region **686**. In an embodiment, the bottom source/drain region (not shown) is connected to power **625** by a backside contact (not shown). In an embodiment, the shared gate region **662**/**682** is connected to cross-couple **690** by frontside contact **692** and the fifth transistor **660**, discussed above. In an embodiment, shared gate region **662**/**682** is the gate region for both the fifth transistor **660** and sixth transistor **680**, discussed above. In an embodiment, the top source/drain region **686** is connected to cross-couple **694** by a frontside contact **688**. In an embodiment, the bottom source/drain region (not shown) may be for the sixth transistor **680**. In an alternative embodiment, the bottom source/drain region (not shown) may be a shared bottom source/drain region that is also a shared bottom source/drain region of any number of sixth transistor(s) in an adjacent SRAM cell(s) (not shown). In an embodiment, as shown in FIG. **6**, the end of fin **684** and the end of the gate region **662**/**682** both extend and align on the RX edge of the sixth transistor **680** closest to the fourth transistor **670**. In an alternative embodiment, the end of fin **684** and the end of gate region **662**/**682** may not align with each other and/or may also not align on the RX edge of the sixth transistor **680**. For example, the edge of fin **684** and gate region **662**/**682** may align on a right edge of the sixth transistor **680** closest to the fifth transistor **660**. In an embodiment, fin **684** has a first length, which is in the same direction as the longer length of SRAM **600** cell, and a second length which is shorter than the first length.
(62) In an embodiment, as discussed above, SRAM **600** cell includes cross-couple **690** and cross-couple **694**. In an embodiment, cross-couple **690** and cross-couple **694** are on any metal layer above the transistors of the SRAM **600** cell. In an embodiment, cross-couple **690** and cross-couple **694** may be a metal wire track. In an embodiment, cross-couple **690** is connected to the first transistor **630**, by frontside contact **638**, third transistor **650** by frontside contact **658**, fourth transistor **670**, by frontside contact **678**, and shared gate region **662**/**682** by frontside contact **692**. In an embodiment, cross-couple **694** is connected to the second transistor **640**, by frontside contact **648**, fifth transistor **660** by frontside contact **668**, sixth transistor **680**, by frontside contact **688**, and shared gate region **652**/**672** by frontside contact **696**.
(63) FIG. **7** depicts a cross-sectional view of section X of an SRAM **700** cell in accordance with an embodiment of the present invention. In an embodiment, as shown in FIG. **7**, SRAM **700** cell includes a backside power delivery network which includes power (i.e., VDD) **725**, ground (i.e., GND) **720**, **722**, bitline **710** and bitline_bar **712**. As discussed above, the backside power delivery network is found on the backside, or between the device layer (i.e., transistors) and the wafer the device layer is formed on. In an embodiment, power **725** may be connected to the backside power delivery network by backside power delivery network contact **725**A. In an embodiment, ground **720**, **722** may be connected to the backside power delivery network by backside power delivery network contact **720**A, **722**A, respectively. In an embodiment, bitline **710** may be connected to the backside power delivery network by backside power delivery network contact **710**A. In an embodiment, bitline_bar **712** may be connected to the backside power delivery network by backside power delivery network contact **712**A.
(64) In an embodiment, bitline **710** is connected to bottom source/drain region **711**. In an embodiment, bitline_bar **712** is connected to bottom source/drain region **713**. In an embodiment, ground **720** is connected to bottom source/drain region **721**. In an embodiment, ground **722** is connected to bottom source/drain region **723**. In an embodiment, power **725** is connected to shared bottom source/drain region **726**. In an alternative embodiment, shared bottom source/drain region **726** may be two individual and separate bottom source/drain regions for each VTFET.
(65) As shown in FIG. **7**, SRAM **700** cell includes a wordline **728**, **729** on either end of the SRAM **700** cell and connected to gate regions **732**, **742**, of the first transistor **630** and second transistor **640**, respectively. In an embodiment, the wordline **728**, **729** are formed on the frontside, or on top of the device layer and on the opposite side the wafer the device layer is formed on. In an

embodiment, the wordlines **728**, **729** is within the traditional back end of the line interconnect wiring. As shown in FIG. **7**, wordlines **728**, **729** are on a first metal layer directly above the transistors of the SRAM **600** cell. In an alternative embodiment, wordline **728**, **729** are on any metal layer above the transistors of the SRAM **600** cell as long as the contact from the wordline **728**, **729** extends to the gate regions **732**, **742**, of the first transistor **630** and second transistor **640**, respectively. In an embodiment, the wordline **728**, **729** is connected to gate regions **732**, **742**, respectively, by a frontside contact.

(66) In an embodiment, gate region **732** surrounds at least a portion of fin **734** and fin **734** is connected to top source/drain region **736**. In an embodiment, top source/drain region **736** is connected to frontside contact **738** and frontside contact **738** is connected to cross-couple **690** (not shown), discussed above. In an embodiment, gate region **742** surrounds at least a portion of fin **744** and fin **744** is connected to top source/drain region **746**. In an embodiment, top source/drain region **746** is connected to frontside contact **748** and frontside contact **748** is connected to cross-couple **694** (not shown), discussed above.

(67) In an embodiment, shared gate region **752**/**772** surrounds at least a portion of fin **754** and fin **774**. In an embodiment, fin **754** is connected to top source/drain region **756**. In an embodiment, top source/drain region **756** is connected to frontside contact **758** and frontside contact **758** is connected to cross-couple **690** (not shown), discussed above. In an embodiment, fin **774** is connected to top source/drain region **776**. In an embodiment, top source/drain region **776** is connected to frontside contact **778** and frontside contact **778** is connected to cross-couple **690** (not shown), discussed above. In an embodiment, shared gate region **752**/**772** is connected to frontside contact **796** and frontside contact **796** is connected to cross-couple **694** (not shown), discussed above.

(68) In an embodiment, shared gate region **762**/**782** surrounds at least a portion of fin **764** and fin **784**. In an embodiment, fin **764** is connected to top source/drain region **766**. In an embodiment, top source/drain region **766** is connected to frontside contact **768** and frontside contact **768** is connected to cross-couple **694** (not shown), discussed above. In an embodiment, fin **784** is connected to top source/drain region **786**. In an embodiment, top source/drain region **786** is connected to frontside contact **788** and frontside contact **788** is connected to cross-couple **694** (not shown), discussed above. In an embodiment, shared gate region **762**/**782** is connected to frontside contact **792** and frontside contact **792** is connected to cross-couple **690** (not shown), discussed above.

(69) FIG. **8** depicts a cross-sectional view of section Y of an SRAM **800** cell in accordance with an embodiment of the present invention. In an embodiment, as shown in FIG. **8**, SRAM **800** cell includes a backside power delivery network which includes power (i.e., VDD) **825**, ground (i.e., GND) **820**, **822**, bitline **810** and bitline_bar **812**. As discussed above, the backside power delivery network is found on the backside, or between the device layer (i.e., transistors) and the wafer the device layer is formed on. In an embodiment, power **825** may be connected to the backside power delivery network by backside power delivery network contact **824**. In an embodiment, ground **820**, **822** may be connected to the backside power delivery network by backside power delivery network contact **819**, **821**, respectively. In an embodiment, bitline **810** may be connected to the backside power delivery network by backside power delivery network contact **809**. In an embodiment, bitline_bar **812** may be connected to the backside power delivery network by backside power delivery network contact **812**A. In an embodiment, power **825** may be connected to the backside power delivery network by backside power delivery network contact **825**A. In an embodiment, ground **820**, **822** may be connected to the backside power delivery network by backside power delivery network contact **820**A, **822**A, respectively. In an embodiment, bitline **810** may be connected to the backside power delivery network by backside power delivery network contact **810**A. In an embodiment, bitline_bar **812** may be connected to the backside power delivery network by backside power delivery network contact **812**A.

(70) In an embodiment, bitline **810** is connected to bottom source/drain region **811**. In an embodiment, bitline_bar **812** is connected to bottom source/drain region **813**. In an embodiment, ground **820** is connected to bottom source/drain region **821**. In an embodiment, ground **822** is connected to bottom source/drain region **823**. In an embodiment, power **825** is connected to shared bottom source/drain region **826**. In an alternative embodiment, shared bottom source/drain region **826** may be two individual and separate bottom source/drain regions for each VTFET.
(71) In an embodiment, as shown in FIG. **8**, SRAM **800** cell includes cross-couple **890**. In an embodiment, cross-couple **890** is connected to frontside contact **838**, frontside contact **858**, frontside contact **878**, and frontside contact **892**. In an embodiment, frontside contact **838** is connected to a top source/drain region (not shown), discussed above. In an embodiment, frontside contact **858** is connected to a top source/drain region (not shown), discussed above. In an embodiment, frontside contact **878** is connected to a top source/drain region (not shown), discussed above. In an embodiment, frontside contact **892** is connected to a shared gate region (not shown), discussed above.
(72) The descriptions of the various embodiments of the present invention have been presented for purposes of illustration but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

Claims

1. A semiconductor memory cell comprising: six vertical-transport field-effect transistors (VTFET) on a wafer, wherein the six VTFET are in a first layer, and wherein the six VTFET are aligned with one another in a first row, wherein each VTFET of the six VTFET includes an isolated fin, and the isolated fin of each VTFET of the six VTFET are aligned with one another in the first row.
2. The semiconductor memory cell of claim 1, wherein power for two of the VTFET of the six VTFET is connected to a power delivery network on a backside of the wafer.
3. The semiconductor memory cell of claim 1, wherein ground for two of the VTFET of the six VTFET is connected to a power delivery network on a backside of the wafer.
4. The semiconductor memory cell of claim 1, wherein a bitline and a bitline bar for two of the VTFET of the six VTFET is connected to a power delivery network on a backside of the wafer and wherein a wordline for two of the VTFET of the six VTFET is connected to a power delivery network on a frontside of the wafer.
5. The semiconductor memory cell of claim 1, wherein a bitline and a bitline bar for two of the VTFET of the six VTFET is connected to a power delivery network on a frontside of the wafer and wherein a wordline for two of the VTFET of the six VTFET is connected to a power delivery network on a backside of the wafer.
6. The semiconductor memory cell of claim 1, wherein a first metal wire electrically connects a source/drain region of a first VTFET, a source/drain region of a second VTFET, a source/drain region of a third VTFET, a gate region of a fourth VTFET, and a gate region of a fifth VTFET.
7. The semiconductor memory cell of claim 6, wherein a second metal wire electrically connects a gate region of the second VTFET, a gate region of the third VTFET, a source/drain region of the fourth VTFET, a source/drain region of the fifth VTFET, and a source/drain region of a sixth VTFET.
8. The semiconductor memory cell of claim 6, wherein the first metal wire is within the semiconductor memory cell.
9. The semiconductor memory cell of claim 7, wherein the second metal wire is within the

semiconductor memory cell.
10. The semiconductor memory cell of claim 1, wherein a fin end and a gate end of one or more VTFET of the six VTFET are aligned to a bottom source/drain region edge.
11. The semiconductor memory cell of claim 1, wherein: the six VTFET form the memory cell; and wherein at least one memory cell is in a row adjacent to at least one other memory cell.
12. A semiconductor SRAM comprising: six vertical-transport field-effect transistors (VTFET) on a wafer, wherein the six VTFET are in a first layer, and wherein the six VTFET are aligned with one another in a first row, wherein each VTFET of the six VTFET includes an isolated fin, and the isolated fin of each VTFET of the six VTFET are aligned with one another in the first row.
13. A semiconductor memory array, comprising: six vertical-transport field-effect transistors (VTFET) aligned with one another and arranged in a memory cell of one or more memory cells in a first layer on a wafer, wherein: each VTFET of the six VTFET includes an isolated fin, and the isolated fin of each VTFET of the six VTFET are aligned with one another in the first row.
14. The semiconductor memory array of claim 13, wherein each memory cell of the one or more memory cells share a first continuous bottom source/drain region for a first VTFET and a second VTFET in each memory cell, wherein the first continuous bottom source/drain region is connected to a power delivery network on a backside of the wafer.
15. The semiconductor memory array of claim 14, wherein each memory cell of the one or more memory cells share a second continuous bottom source/drain region for a third VTFET and a fourth VTFET in each memory cell, wherein the second continuous bottom source/drain region is connected to a power delivery network on the backside of the wafer.
16. The semiconductor memory array of claim 14, wherein each memory cell of the one or more memory cells share a third continuous bottom source/drain region for a fifth VTFET and a fourth continuous bottom source/drain region for a sixth VTFET in each memory cell, wherein the third continuous bottom source/drain region is connected to a power delivery network on a backside of the wafer, and wherein the fourth continuous bottom source/drain region is connected to a power delivery network on the backside of the wafer.
17. The semiconductor memory array of claim 13, wherein each memory cell of the one or more memory cells are arranged in a single row.